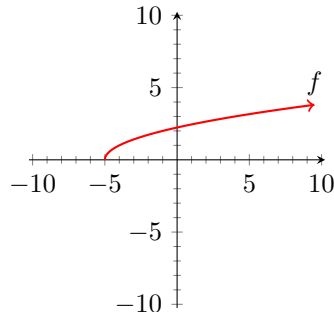


Justify your answers and show your work. Unless indicated otherwise, each problem is worth 5 points.

1. Let $f(x) = \sqrt{x+5}$ for $x \geq -5$ [3 pts each]

a. Find the domain of f^{-1} .

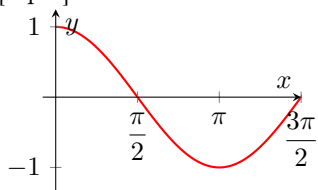
- c. Graph f^{-1} .



b. Find the range of f^{-1} .

1. Using the graph of $f(x)$ shown below to complete the following.

a. [4 pts] Find two intervals on which f has an inverse, making each interval as large as possible.



b. [2 pts] Assuming that f^{-1} exists on one of these intervals, find $f^{-1}(1)$.

2. Given the function $f(x) = (x-2)^2$ for $x \geq 2$, find the slope of the line tangent to the graph of f^{-1} at the point $(1, 3)$.

Perform the indicated operation.

3. $\frac{d}{dx}(e^{-3x} + 5e^x)$

4. $\frac{d}{dx}x^2 \ln x - \pi$

Perform the indicated operation.

5. $\int \left(10e^{5x} + \frac{1}{x} \right) dx$

6. $\int \frac{dx}{3x - 5}$

7. Use logarithmic differentiation to find the derivative of $f(x) = (1 + e^x)^x$.

8. Solve the equation for x .

a. [3 pts] $e^{2x} = 30$

b. [3 pts] $\ln 2x = 5$

9. $\frac{d}{dx} \log_3 x$

10. $\frac{d}{dx} \frac{3^x}{x^2}$

Perform the indicated operation.

11. $\frac{d}{dx} (x^\pi + 1)$

12. [8 pts] $\int 2xe^{2x} dx$

13. [8 pts] $\int x \cos(8x) dx$

14. [8 pts] $\int 10x \ln(2x) dx$

Bonus:

15. $\int x \cdot \sin x \cos x dx$